

LAB ASSIGNMENT 6

PROBLEM 1: Apply the KNN Algorithm on all three given datasets and analyze their performance.

CODE:

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier

k = int(input("Enter value of K: "))
df_diabetes = pd.read_csv('diabetes_dataset.csv')
print("Diabetes")
print(df_diabetes.head())
print("Missing values")
print(df_diabetes.isnull().sum())
df_social = pd.read_csv('Social_Network_Ads.csv')
print("\nSocial Network Ads")
print(df_social.head())
print("Missing values")
print(df_social.isnull().sum())

df_titanic = pd.read_csv('titanic.csv')
print("\nTitanic")
print(df_titanic.head())
print("Missing values")
print(df_titanic.isnull().sum())

df_titanic['Age'] =
df_titanic['Age'].fillna(df_titanic['Age'].median())
df_titanic['Embarked'] =
df_titanic['Embarked'].fillna(df_titanic['Embarked'].mode()[0])
df_titanic = df_titanic.drop(columns=['Cabin'])

print("Missing values (after analysis)")
print(df_titanic.isnull().sum())
for col in ['Glucose', 'BloodPressure', 'SkinThickness', 'Insulin',
            'BMI']:
```

```

df_diabetes[col] = df_diabetes[col].replace(0,
df_diabetes[col].median())

X_d = StandardScaler().fit_transform(df_diabetes.drop('Outcome',
axis=1))
y_d = df_diabetes['Outcome'].values
X_tr_d, X_te_d, y_tr_d, y_te_d = train_test_split(X_d, y_d,
test_size=0.2, random_state=42)

df_social['Gender'] =
LabelEncoder().fit_transform(df_social['Gender'])
X_s = StandardScaler().fit_transform(df_social.drop(['User ID',
'Purchased'], axis=1))
y_s = df_social['Purchased'].values
X_tr_s, X_te_s, y_tr_s, y_te_s = train_test_split(X_s, y_s,
test_size=0.2, random_state=42)

df_titanic = df_titanic.drop(columns=['PassengerId', 'Name',
'Ticket'])
df_titanic['Sex'] = LabelEncoder().fit_transform(df_titanic['Sex'])
df_titanic['Embarked'] =
LabelEncoder().fit_transform(df_titanic['Embarked'])
X_t = StandardScaler().fit_transform(df_titanic.drop('Survived',
axis=1))
y_t = df_titanic['Survived'].values
X_tr_t, X_te_t, y_tr_t, y_te_t = train_test_split(X_t, y_t,
test_size=0.2, random_state=42)

datasets = [
    ("Diabetes", X_tr_d, X_te_d, y_tr_d, y_te_d),
    ("Social Ads", X_tr_s, X_te_s, y_tr_s, y_te_s),
    ("Titanic", X_tr_t, X_te_t, y_tr_t, y_te_t)
]

for name, X_tr, X_te, y_tr, y_te in datasets:
    print(f"\n{name} Results")
    model = KNeighborsClassifier(n_neighbors=k)
    model.fit(X_tr, y_tr)
    acc = model.score(X_te, y_te)
    print(f"Accuracy: {acc:.4f}")

```

OUTPUT:

```

PS D:\NITJ ACADEMICS\OPENLEARN> python -u "d:\NITJ ACADEMICS\OPENLEARN\LAB_2026\LAB_7_2.py"
Enter value of K: 3
Diabetes
  Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin  BMI  DiabetesPedigreeFunction  Age  Outcome
0             6      148             72             35         0   33.6              0.627   50       1
1             1       85             66             29         0   26.6              0.351   31       0
2             8      183             64             0          0   23.3              0.672   32       1
3             1       89             66             23        94   28.1              0.167   21       0
4             0      137             40             35       168   43.1              2.288   33       1
Missing values
Pregnancies           0
Glucose               0
BloodPressure         0
SkinThickness         0
Insulin               0
BMI                   0
DiabetesPedigreeFunction 0
Age                   0
Outcome               0
dtype: int64

Social Network Ads
  User ID  Gender  Age  EstimatedSalary  Purchased
0  15624510   Male   19           19000         0
1  15810944   Male   35           20000         0
2  15668575  Female   26           43000         0
3  15603246  Female   27           57000         0
4  15804002   Male   19           76000         0
Missing values
User ID           0
Gender            0
Age               0
EstimatedSalary   0
Purchased         0
dtype: int64

Titanic
  PassengerId  Survived  Pclass                               Name  Sex  Age  SibSp  Parch  Ticket  Fare  Cabin  Embarked
0             1         0       3                Braund, Mr. Owen Harris  male  22.0    1     0    A/5 21171   7.2500   NaN     S
1             2         1       1  Cumings, Mrs. John Bradley (Florence Briggs Th... female  38.0    1     0    PC 17599  71.2833   C85     C
2             3         1       3      Heikkinen, Miss. Laina          female  26.0    0     0  STON/O2. 3101282   7.9250   NaN     S
3             4         1       1  Futrelle, Mrs. Jacques Heath (Lily May Peel) female  35.0    1     0    113803  53.1000  C123     S
4             5         0       3      Allen, Mr. William Henry          male  35.0    0     0    373450   8.0500   NaN     S
Missing values
PassengerId       0
Survived          0
Pclass            0
Name              0
Sex               0
Age              177
SibSp             0
Parch             0
Ticket            0
Fare              0
Cabin            687
Embarked          2
dtype: int64
Missing values (after analysis)
PassengerId       0
Survived          0
Pclass            0
Name              0
Sex               0
Age              0
SibSp             0
Parch             0
Ticket            0
Fare              0
Embarked          0
dtype: int64

Diabetes Results
Accuracy: 0.7208

Social Ads Results
Accuracy: 0.9125

Titanic Results
Accuracy: 0.7877

```

PROBLEM 2: Apply the KNN Algorithm on all three given datasets and analyze their performance. (HARDCODE)

CODE:

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.model_selection import train_test_split
from scipy.spatial.distance import euclidean, cityblock

k = int(input("Enter value of K: "))
df_d = pd.read_csv('diabetes_dataset.csv')
print("Diabetes")
print(df_d.head())
print("Missing values")
print(df_d.isnull().sum())

df_s = pd.read_csv('Social_Network_Ads.csv')
print("\nSocial Network Ads")
print(df_s.head())
print("Missing values")
print(df_s.isnull().sum())

df_t = pd.read_csv('titanic.csv')
print("\nTitanic")
print(df_t.head())
print("Missing values")
print(df_t.isnull().sum())

df_t['Age'] = df_t['Age'].fillna(df_t['Age'].median())
df_t['Embarked'] =
df_t['Embarked'].fillna(df_t['Embarked'].mode()[0])
df_t = df_t.drop(columns=['Cabin'])

print("Missing values.")
print(df_t.isnull().sum())
def dtw_dist(s, t):
    n, m = len(s), len(t)
```

```

    dtw = np.zeros((n + 1, m + 1))
    dtw[1:, 0] = np.inf
    dtw[0, 1:] = np.inf
    for i in range(1, n + 1):
        for j in range(1, m + 1):
            c = abs(s[i-1] - t[j-1])
            dtw[i, j] = c + min(dtw[i-1, j], dtw[i, j-1], dtw[i-1,
j-1])
    return dtw[n, m]
def knn_pred(X_tr, y_tr, X_te, k, m):
    preds = []
    for tp in X_te:
        dists = []
        for tr in X_tr:
            if m == 'L1':
                d = cityblock(tp, tr)
            elif m == 'L2':
                d = euclidean(tp, tr)
            elif m == 'DTW':
                d = dtw_dist(tp, tr)
            dists.append(d)
        idx = np.argsort(dists)[:k]
        preds.append(np.bincount(y_tr[idx]).argmax())
    return np.array(preds)

for col in ['Glucose', 'BloodPressure', 'SkinThickness', 'Insulin',
'BMI']:
    df_d[col] = df_d[col].replace(0, df_d[col].median())
X_d = StandardScaler().fit_transform(df_d.drop('Outcome', axis=1))
y_d = df_d['Outcome'].values
X_tr_d, X_te_d, y_tr_d, y_te_d = train_test_split(X_d, y_d,
test_size=0.2, random_state=42)
df_s['Gender'] = LabelEncoder().fit_transform(df_s['Gender'])
X_s = StandardScaler().fit_transform(df_s.drop(['User ID',
'Purchased'], axis=1))
y_s = df_s['Purchased'].values
X_tr_s, X_te_s, y_tr_s, y_te_s = train_test_split(X_s, y_s,
test_size=0.2, random_state=42)

df_t = df_t.drop(columns=['PassengerId', 'Name', 'Ticket'])
df_t['Sex'] = LabelEncoder().fit_transform(df_t['Sex'])
df_t['Embarked'] = LabelEncoder().fit_transform(df_t['Embarked'])

```

```

X_t = StandardScaler().fit_transform(df_t.drop('Survived', axis=1))
y_t = df_t['Survived'].values
X_tr_t, X_te_t, y_tr_t, y_te_t = train_test_split(X_t, y_t,
test_size=0.2, random_state=42)
data = [
    ("Diabetes", X_tr_d, X_te_d, y_tr_d, y_te_d),
    ("Social Ads", X_tr_s, X_te_s, y_tr_s, y_te_s),
    ("Titanic", X_tr_t, X_te_t, y_tr_t, y_te_t)
]
for name, X_tr, X_te, y_tr, y_te in data:
    print(f"\n{name} Results")
    for m in ['L1', 'L2', 'DTW']:
        y_p = knn_pred(X_tr, y_tr, X_te, k, m)
        acc = (y_p == y_te).mean()
        print(f"{m} Accuracy: {acc:.4f}")

```

OUTPUT:

```

PS D:\NITJ ACADEMICS\OPENLEARN> python -u "d:\NITJ ACADEMICS\OPENLEARN\LAB_2026\LAB_7_1.py"
Enter value of K: 5
Diabetes

```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1
3	1	89	66	23	94	28.1	0.167	21	0
4	0	137	40	35	168	43.1	2.288	33	1

```

Missing values
Pregnancies      0
Glucose           0
BloodPressure     0
SkinThickness     0
Insulin           0
BMI               0
DiabetesPedigreeFunction 0
Age               0
Outcome           0
dtype: int64

Social Network Ads

```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0

```

Missing values
User ID           0
Gender            0
Age               0
EstimatedSalary   0
Purchased         0
dtype: int64

```

```

Titanic
  PassengerId  Survived  Pclass
0             1         0       3
1             2         1       1  Cumings, Mrs. John Bradley (Florence Briggs Th...
2             3         1       3                Heikkinen, Miss. Laina
3             4         1       1  Futrelle, Mrs. Jacques Heath (Lily May Peel)
4             5         0       3                Allen, Mr. William Henry
Missing values
PassengerId    0
Survived        0
Pclass          0
Name            0
Sex             0
Age            177
SibSp           0
Parch           0
Ticket          0
Fare            0
Cabin          687
Embarked        2
dtype: int64
Missing values.
PassengerId    0
Survived        0
Pclass          0
Name            0
Sex             0
Age             0
SibSp           0
Parch           0
Ticket          0
Fare            0
Embarked        0
dtype: int64

Diabetes Results
L1 Accuracy: 0.6948
L2 Accuracy: 0.7468
DTW Accuracy: 0.7403

Social Ads Results
L1 Accuracy: 0.9250
L2 Accuracy: 0.9250
DTW Accuracy: 0.8750

Titanic Results
L1 Accuracy: 0.8268
L2 Accuracy: 0.7989
DTW Accuracy: 0.8324

```